Reinforcement Learning for Robust Quadrupedal Locomotion: Training a Unitree Go2 with PPO in Isaac Lab

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Repo: https://github.com/Adgonzalez2018/rob6323_go2_project

***Abstract*—We present a comprehensive approach to training robust quadrupedal locomotion for the Unitree Go2 robot using Proximal Policy Optimization (PPO) in Isaac Lab. Starting from a minimal baseline that only rewards velocity tracking, we systematically improved gait quality through principled reward engineering. We successfully implemented two bonus tasks: (1) an actuator friction model with domain randomization ($\mu_v \sim U(0, 0.3)$, $F_s \sim U(0, 2.5)$) that maintains stable gaits despite reduced performance, and (2) push recovery training with velocity impulses demonstrating robust disturbance rejection. Our baseline achieves ∼1250 total reward with smooth trotting gaits. The friction-adapted policy maintains functionality at ∼1240 reward, while push recovery configurations vary significantly: mild pushes (10-30N) achieve ∼625 reward with full episodes, while aggressive pushes (15-60N) cause early terminations at ∼33 reward.**

I. Introduction

Legged locomotion represents one of the most challenging problems in robotics, requiring simultaneous optimization of stability, efficiency, and adaptability to environmental uncertainties. Recent advances in deep reinforcement learning (RL) have shown promise in learning robust locomotion policies directly from experience [1], [2].

In this project, we train a quadrupedal locomotion policy for the Unitree Go2 robot using Proximal Policy Optimization (PPO) in Isaac Lab. Our objectives are threefold: (1) develop a baseline policy that exhibits stable, periodic gaits while accurately tracking velocity commands, (2) adapt the policy to realistic actuator dynamics through friction modeling, and (3) enhance robustness to external disturbances through push recovery training.

II. Methodology

A. Environment and Training Setup

We utilize Isaac Lab's physics simulation with the Unitree Go2 robot model, featuring 12 degrees of freedom. The simulation runs at 200 Hz with a control frequency of 50 Hz (decimation factor of 4). Each training episode lasts 20 seconds.

The observation space consists of 52 dimensions including proprioceptive information (joint positions/velocities), base state, velocity commands, and gait timing clocks. The action space comprises 12 dimensions representing target joint angles, scaled by 0.25 radians.

B. Reward Engineering

Our reward function combines multiple objectives:

$$R_{total} = R_{tracking} + R_{stability} + R_{gait} + R_{reg} \tag{1}$$

Velocity Tracking: We use an exponential kernel:

$$R_{vel} = \exp\left(-\frac{\|\mathbf{v}_{cmd} - \mathbf{v}_{actual}\|^2}{\sigma^2}\right) \tag{2}$$

Stability Rewards:

- Orientation: $R_{orient} = -5.0 \cdot \|\mathbf{g}_{xy}\|^2$
- Vertical velocity: $R_{vz} = -0.02 \cdot v_z^2$
- Angular velocity: $R_\omega = -0.001 \cdot \|\boldsymbol{\omega}_{xy}\|^2$

Raibert Heuristic: For gait quality:

$$\mathbf{p}_{foot}^{desired} = \mathbf{p}_{nom} + \phi \cdot \mathbf{v}_{cmd} \cdot \frac{T_{stance}}{2f} \tag{3}$$

Action Regularization:

- Action rate: $R_{rate} = -0.1 \cdot (\|\mathbf{a}_t - \mathbf{a}_{t-1}\|^2)$
- Joint velocity: $R_{\dot{q}} = -0.0001 \cdot \|\dot{\mathbf{q}}\|^2$
- Torque: $R_\tau = -0.00001 \cdot \|\boldsymbol{\tau}\|^2$

C. Custom PD Controller

We implement explicit PD control:

$$\boldsymbol{\tau} = \text{clip}(K_p(\mathbf{q}_d - \mathbf{q}) - K_d\dot{\mathbf{q}}, -\tau_{max}, \tau_{max}) \tag{4}$$

with $K_p = 20.0$, $K_d = 0.5$, and $\tau_{max} = 100$ Nm.

III. Implementation

A. Baseline Development

Our baseline implementation progressed through five key improvements:

1) **Action Rate Penalties:** 3-step history buffer penalizing derivatives
2) **Custom PD Controller:** Disabled implicit control for transparency
3) **Early Termination:** Added conditions for failed states
4) **Raibert Heuristic:** Guided foot placement with phase clocks
5) **Stability Rewards:** Comprehensive posture penalties

B. Bonus Task 1: Actuator Friction Model

Following [2], we model actuator friction as:

$$\tau_{friction} = F_s \cdot \tanh\left(\frac{\dot{q}}{0.1}\right) + \mu_v \cdot \dot{q} \tag{5}$$

with domain randomization per episode:

$$\mu_v \sim \mathcal{U}(0.0, 0.3) \tag{6}$$
$$F_s \sim \mathcal{U}(0.0, 2.5) \tag{7}$$

C. Bonus Task 2: Push Recovery

External disturbances through velocity impulses:

$$\Delta\mathbf{v} = \frac{F}{m_{scale}} \cdot [\cos(\theta), \sin(\theta), 0]^T \tag{8}$$

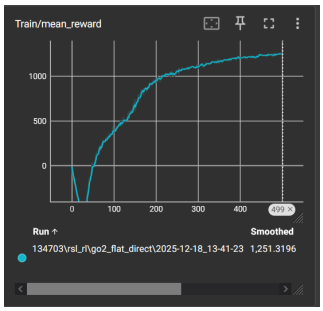
We tested two configurations:

- **Mild:** $F \in [10, 30]$N, interval = 50 steps
- **Aggressive:** $F \in [15, 60]$N, interval = 50 steps

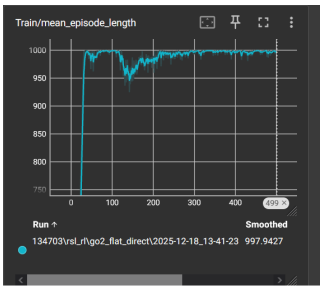
IV. Results and Analysis

A. Training Performance

Figure 1 shows the baseline training curves, achieving convergence at ∼1250 mean reward with stable episode lengths near the maximum of 1000 steps.



(a) Baseline reward



(b) Baseline episode length

Fig. 1: Baseline training performance showing stable convergence at ∼1250 reward with consistent full episode completion.

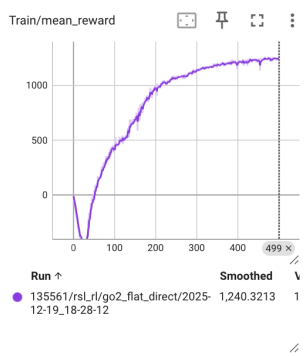
B. Friction Model Adaptation

The friction model training (Fig. 2) maintains comparable performance despite added resistance, converging at ∼1240 reward. This minimal degradation validates successful adaptation to actuator dynamics.

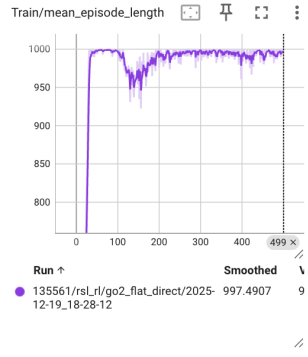
C. Push Recovery Performance

Push recovery results reveal a critical trade-off between robustness and stability, as shown in Fig. 3.

The mild configuration (10-30N) achieves optimal balance with ∼625 reward while maintaining full episode lengths. In contrast, aggressive pushes (15-60N) cause catastrophic failures with episodes terminating at ∼137 steps and rewards of only ∼33.

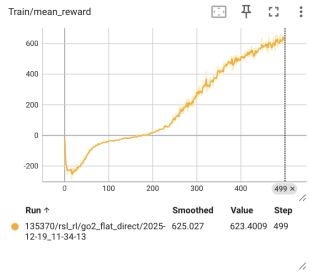


(a) Friction reward

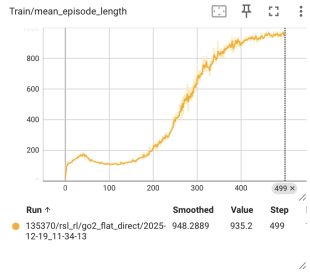


(b) Friction episode length

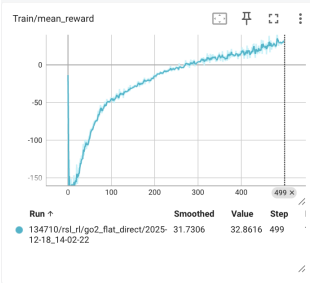
Fig. 2: Friction model training showing robust adaptation with minimal performance degradation (∼1240 reward).



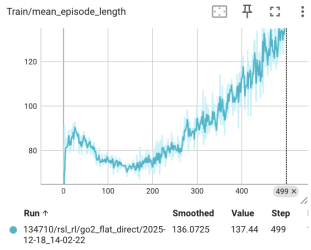
(a) Mild push reward



(b) Mild push length



(c) Aggressive reward



(d) Aggressive length

Fig. 3: Push recovery training: (a-b) Mild pushes maintain full episodes with ∼625 reward; (c-d) Aggressive pushes cause early terminations at ∼137 steps.

TABLE I: Performance Comparison Across Configurations

Config	Reward	Ep. Length	Stability
Baseline	1251	998	Excellent
Friction	1240	997	Excellent
Push (Mild)	625	935	Good
Push (Aggr.)	33	137	Poor

D. Comparative Analysis

Table I summarizes performance across all configurations.

V. Discussion

A. Key Findings

Reward Engineering Criticality: Progressive reward shaping proved essential for gait quality. The Raibert heuristic effectively established periodic foot patterns.

Robustness-Performance Trade-offs: Both friction and push recovery reduce tracking performance while enhancing real-world applicability. Yaw control consistently remained robust across all configurations.

Optimal Disturbance Magnitude: Mild pushes (10-30N) provide sufficient challenge for learning recovery without inducing constant failures, unlike aggressive pushes that overwhelm the policy.

B. Limitations and Future Work

Terrain Generalization: Failed attempts revealed reward exploitation where robots maximize reward by standing still. Future work requires explicit forward progress rewards and curriculum learning.

Sim-to-Real Gap: Additional factors for real deployment include sensor noise, ground contact dynamics, and motor temperature effects.

VI. Conclusion

We successfully trained robust quadrupedal locomotion policies using PPO in Isaac Lab. Our baseline achieves $\sim$1250 reward with stable trotting gaits. The friction model maintains comparable performance ($\sim$1240) despite actuator resistance. Push recovery reveals critical insights: mild disturbances (10-30N) enable robust learning with $\sim$625 reward, while aggressive pushes (15-60N) cause catastrophic failures.

Key contributions include: (1) systematic reward engineering for quality gaits, (2) successful friction adaptation with minimal degradation, (3) identification of optimal disturbance ranges for robustness training, and (4) insights into reward exploitation in terrain generalization.

This work provides a foundation for real-world deployment, with friction-adapted policies being strong candidates for hardware testing.

References

[1] E. Chane-Sane et al., "SoloParkour: Constrained Reinforcement Learning for Visual Locomotion," CoRL, 2024.

[2] J. Amigo et al., "First Order Model-Based RL Through Decoupled Backpropagation," CoRL, 2025.

[3] J. Schulman et al., "Proximal Policy Optimization Algorithms," arXiv:1707.06347, 2017.

[4] M. H. Raibert, *Legged Robots That Balance*. MIT Press, 1986.

[5] Isaac Lab Team, "Isaac Lab Documentation," 2024. [Online]. Available: https://isaac-sim.github.io/IsaacLab/